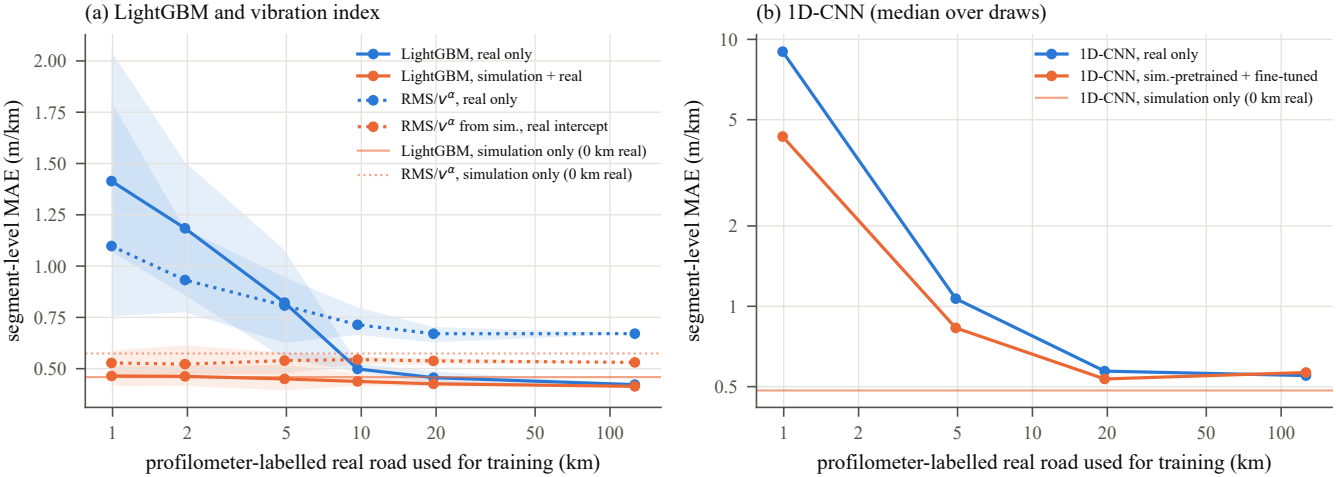
(a) LightGBM and vibration index
segment-level MAE (m/km)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
LightGBM, real only
LightGBM, simulation + real
RMS/v^{α}, real only
RMS/v^{α} from sim., real intercept
LightGBM, simulation only (0 km real)
RMS/v^{α}, simulation only (0 km real)
1
2
5
10
20
50
100
profilometer-labelled real road used for training (km)
(b) 1D-CNN (median over draws)
segment-level MAE (m/km)
10
5
2
1
0.5
1D-CNN, real only
1D-CNN, sim.-pretrained + fine-tuned
1D-CNN, simulation only (0 km real)
1
2
5
10
20
50
100
profilometer-labelled real road used for training (km)